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Quiz 8

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Are the following arguments valid?

1)

If I drive to work, then I will arrive tired.
I am not tired when I arrive at work.

p is drive to work, q is arrive tired
 $p \rightarrow q \quad \neg q \rightarrow \neg p$

Therefore I do not drive to work

valid

2)

If I drive to work, then I will arrive tired.
I do not drive to work.

p is drive to work, q is arrive tired
 $p \rightarrow q \quad \neg p \rightarrow \neg q$

Therefore I will not arrive tired

invalid

p	q	$p \rightarrow q$	$\neg p$	$p \rightarrow q \wedge \neg p$	$\neg q$	Q
0	0	1	1	1	1	1
0	1	1	1	1	0	1
1	0	0	0	0	1	0
1	1	1	0	0	0	1

3)

I will become famous or I will not become a writer.
I will become a writer

p is become famous, q become a writer
 $p \vee \neg q \quad \neg q \rightarrow p$

Therefore I will become famous.

valid

p	q	$\neg q$	$p \vee \neg q$	$p \vee \neg q \wedge \neg q$	Q
0	0	1	1	0	1
0	1	0	0	0	1
1	0	1	1	0	1
1	1	0	1	0	1

4)

Prove the sum of an even integer and an odd integer is odd

let m be even and n be odd
so $m = 2k$ and $n = 2k+1$
so $m+n = 2k + 2k+1 = 4k+1$
so $4k+1 = 2(2k)+1$, is odd

5)

Prove if $2y$ is irrational, then y is irrational

if y is rational, then $2y$ is rational

so $y = \frac{a}{b}$, so $2y = \frac{2a}{b}$

this $2y$ is rational.